

IMPERIAL

Backtest of ICES Advice

Initial results and questions for discussion

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Agenda

- 01** Why a backtest
- 02** Case-study stocks
- 03** What we have done
- 04** Initial results
- 05** Questions for you
- 06** Next steps

Two questions that are often mixed

Why this project

Was the science robust?

Given the assessment and harvest control rule in force at the time, would following advice have delivered the intended status, yield and stability?

Was advice followed?

Did realised catch match advised catch? If not, how much of the poor outcome is implementation, not the rule itself?

A **backtest** is a retrospective management strategy evaluation: reconstruct each year's perception, apply that year's rule, project under perfect implementation, and compare with what actually happened.

What success looks like

For management, not only for the model

- Separate **implementation failure** from **advice-rule failure**.
- Show, stock by stock, whether following advice would have rebuilt biomass, reduced overfishing, or mainly changed yield in the short term.
- Identify where the ICES MSY rule itself is the constraint (zero-catch advice, unstable assessments, benchmark breaks).
- Leave a reproducible workflow that can be updated as new assessments arrive.

Six Category 1 case studies

All retained after week-1 screening

Stock	Area	Assessment	Advice basis
Irish Sea cod	7.a	Age-structured	ICES MSY
Irish Sea whiting	7.a	SAM	ICES MSY
Celtic Sea cod	7.e-k	SAM	ICES MSY
Celtic Sea whiting	7.b-c, 7.e-k	Age-based	ICES MSY
Celtic Sea haddock	7.b-k	SAM	ICES MSY
NE Atlantic mackerel	1-8, 14, 9.a	Integrated	Coastal States MSY

SAG coverage is typically 2016/17–2025/26 (about 8–10 assessment years). Irish Sea whiting is the sparsest; mackerel is retained **provisionally**.

How the backtest works

Repeated for each historical assessment year t

1. **Perception** — reconstruct status using only data and reference points available at t .
2. **Advice** — apply the harvest control rule then in force ($F = F_{\text{MSY}}$ above $\text{MSYB}_{\text{trigger}}$; linear reduction to zero at B_{lim}).
3. **Counterfactual** — project the operating model with catch equal to advised catch (perfect implementation).
4. **Compare** — SSB, F , catch, time below $B_{\text{lim}}/\text{MSYB}_{\text{trigger}}$, versus realised history.

Where we are

Contract workplan: 50 days

Task	Content	Days	Status
1	Screening, status, catch vs advice, open-loop F checks	5	Done
2	Assessment reconstruction (historical peels)	20	Next
3	Operating models and perfect-management HCR runs	15	Scaffolded
4	Metrics, synthesis, technical report	10	Draft shell

Pipeline, LaTeX report, and ICES HCR code are in place. Remaining work is running, checking, and writing — not starting from a blank sheet.

Initial results from screening

Latest status is poor for most stocks

SAG terminal year relative to current reference points

Stock	SSB/MSYB _{trigger}	SSB/B _{lim}	F/F _{MSY}
Irish Sea cod	0.28	0.38	0.05
Irish Sea whiting	0.63	0.87	—
Celtic Sea cod	0.02	0.02	4.95
Celtic Sea whiting	0.26	0.36	0.65
Celtic Sea haddock	0.63	0.88	1.78
NE Atlantic mackerel	0.77	1.03	1.39

- All six are below MSYB_{trigger}; Celtic Sea cod is far below B_{lim}.
- Celtic Sea cod, haddock and mackerel: $F > F_{MSY}$ in the latest year.
- Irish Sea cod: F already low, but biomass has not rebuilt — weak year classes since ~2020.

Advice was often not followed

Reported catch versus ICES advised catch

- **Celtic and Irish Sea cod and whiting:** advised catch cuts were not fully taken.
- **Mackerel:** catch has consistently exceeded advice.
- **Celtic Sea haddock:** catch has generally been below advice — a different kind of mismatch.

That gap is why a backtest is informative. If advice had always been followed, we could only test the science, not management effectiveness.

Screening uses catch versus advice only — not TAC. Perfect-management runs assume catch = advised catch.

Constant-F checks: fishing vs recruitment

Open-loop screening projections, not the closed-loop backtest

Three tracks at the same target $F = F_{MSY}$:

- **Actual** — realised F and recruitment.
- **Mean recruitment** — isolates a deterministic F effect.
- **Recruitment deviates** — same F , historical year-class pattern.

What this already suggests

- Celtic Sea cod: $F \gg F_{MSY}$, $SSB \ll B_{lim}$ — following a lower F is the clearest screening case for higher biomass.
- Celtic Sea whiting: biomass poor but $F < F_{MSY}$; recruitment, not only F , is driving the path.
- Irish Sea cod: low F has not rebuilt the stock; the counterfactual is not “just fish less”.

Assessments move: updates vs benchmarks

Perception of status is not a fixed time series

- Update years mainly revise the most recent points (usual retrospective fan).
- Benchmarks can change **scale and reference points** — mackerel, Celtic Sea whiting, Irish Sea cod.
- For mackerel the recent benchmark shifted both the stock trend and the reference-point basis. Screening numbers for that stock are **provisional**.

The backtest must use **the reference points in force in year t** , not only today's F_{MSY} and $MSYB_{trigger}$.

Screening decision

What we have already taken as given

- **Proceed with all six stocks.** No alternative list proposed.
- Mackerel kept, but benchmark handling must be agreed before full conditioning.
- Operating-model biology can follow the assessment; the critical choice is the **stock-recruitment relationship** and how deviates are used in projection.
- Short-term reference projections from the latest assessment (as in the project plan) are still outstanding — distinct from the whole-period screening runs.

Questions where we need clarity

Q1 --- Mackerel

Retain, special-case, or drop?

The recent benchmark changed perceived scale and reference points, not only the last few years.

What I need from you

- Keep it as a full sixth case study?
- Keep it, but freeze pre-benchmark reference points / scale with an agreed rule?
- Park it until the other five are done, then return?

This is the highest-risk stock in the set. Getting the rule wrong will dominate the narrative.

Q2 --- How far back do we peel?

Assessment window

Practical window is about **8–10 years** (2016/17–2025/26).

What I need from you

- Is that long enough for the decision you want to support?
- Prefer a common window across stocks, or the maximum SAG years per stock?
- Irish Sea whiting has gaps and recent zero-catch advice — same window, or a shorter documented subset?

Q3 --- Reconstructing historical assessments

SAG gives outputs, not the inputs as they stood in year t

We cannot fully rerun “the assessment as published” from SAG archives alone.

Options

1. **Simulation:** hold catch-at-age and biology; simulate CPUE; rerun SAM — closer to a true peel, more work.
2. **Approximate peels** from the latest assessment — faster, weaker claim.
3. **Hybrid:** simulation on one or two stocks; approximation on the rest.

Which claim do you need the report to be able to make?

Q4 --- What does “perfect management” mean?

The counterfactual definition

Screening compared **catch vs advised catch**. TAC is not in the figures yet.

What I need from you

- Is the policy question “what if catch had equalled ICES advice”?
- Or do we also need TAC vs advice vs catch (three-way mismatch)?
- For years with **zero-catch advice** (Irish Sea whiting), is the intended story “perfect management means no fishing”, even if that makes short-term yield look worse by design?

Q5 --- What decision should the report serve?

This shapes metrics, depth, and how we spend the remaining days

Possible uses (not mutually exclusive):

- Show managers the cost of not following advice.
- Test whether the ICES MSY rule would have rebuilt these stocks if followed.
- Inform HCR / long-term management plan discussions (including rebuilding).
- A technical methods report, with policy implications secondary.

Which of these is primary for you?

Q6 --- Sequencing and depth

About 45 days remain

Proposal: one stock end-to-end first (Celtic Sea cod is the clearest screening case), then roll the same workflow across the others.

What I need from you

- Agree Celtic Sea cod as the prototype, or nominate another?
- Equal depth on all six, or a full treatment plus lighter companions?
- Final package: technical report only, or also a short briefing note / these slides updated with counterfactuals?

What I need from this meeting

A usable decision on each

Decision

- Q1 Mackerel: full case / special rule / defer
 - Q2 Peel window: common years vs maximum per stock
 - Q3 Reconstruction: simulation, approximation, or hybrid
 - Q4 Counterfactual: catch = advice; TAC or not; zero-catch years
 - Q5 Primary use of the report
 - Q6 Prototype stock and depth vs breadth
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Immediate next steps (after today)

1. Condition the operating model for the prototype stock and run open- and closed-loop tests.
2. Apply the agreed peel method and contemporaneous HCR (year-specific reference points).
3. Repeat across the confirmed list; treat mackerel as agreed in Q1.
4. Fill the technical report from those runs (outline already matches the contract markup).

Screening PDF and confirmed stock list are already a contract deliverable.